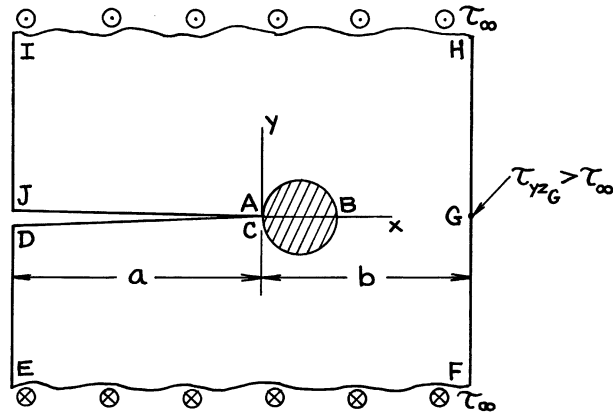
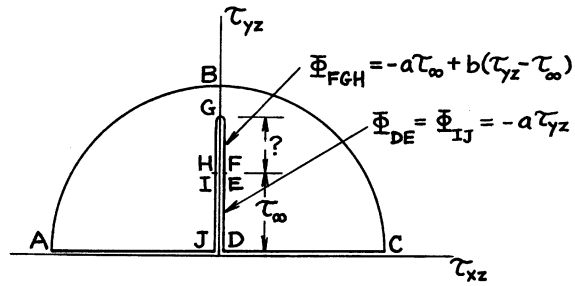


Physical Plane



Stress Plane



Sketch of Stress Function Surface

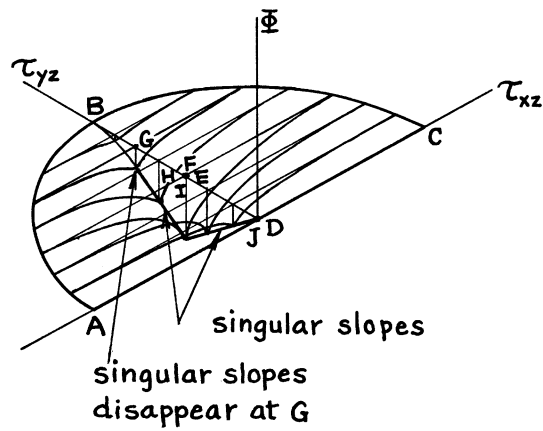
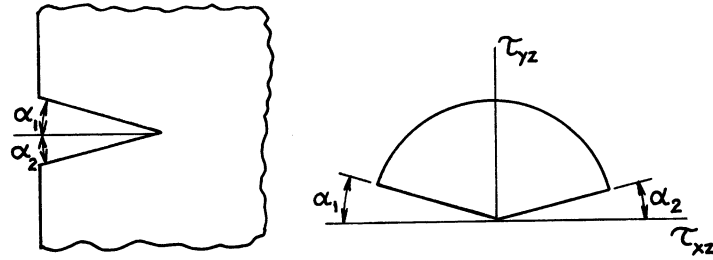
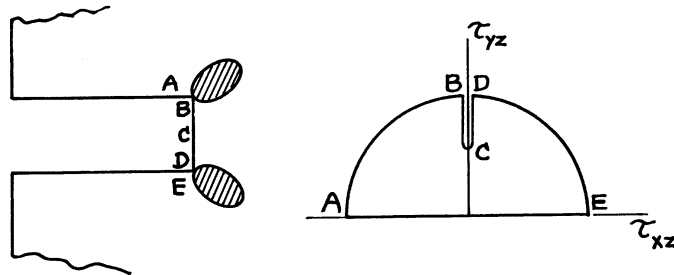


Fig. K.6

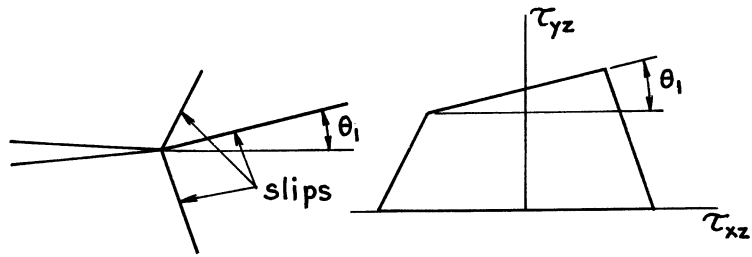
Effect of Angled Sharp Notches:



Effect of Square Ended Notches:



Effect of Discrete Limited Slip Directions:



For Plane Anisotropic Elastic Behavior:

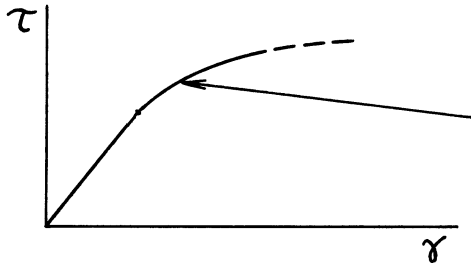
$$\tau_{xz} = A_{xx} \gamma_{xz} + A_{xy} \gamma_{yz}$$

$$\tau_{yz} = A_{yx} \gamma_{xz} + A_{yy} \gamma_{yz} \quad \text{with} \quad A_{yx} = A_{xy}$$

results in replacing Laplace's Equation for the elastic region by

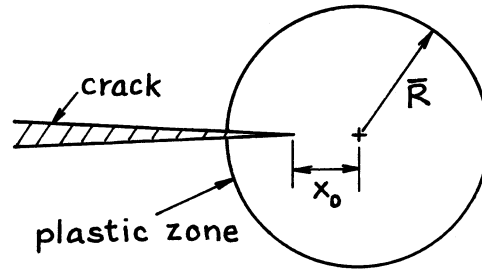
$$A_{xx} \frac{\partial^2 \Phi}{\partial \tau_{xz}^2} + 2A_{xy} \frac{\partial^2 \Phi}{\partial \tau_{xz} \partial \tau_{yz}} + A_{yy} \frac{\partial^2 \Phi}{\partial \tau_{yz}^2} = 0$$

For Power Hardening Plasticity Beyond an Elastic Range:



$$\frac{\tau}{\tau_0} = \left(\frac{\gamma}{\gamma_0} \right)^N$$

$$x_0 = \frac{1-N}{1+N} \bar{R}$$



(See **Rice 1968c** for details and numerical solutions.)

APPENDIX L

TABLE OF COMPLETE ELLIPTIC INTEGRALS, $K(k)$ AND $E(k)$

$$k = \sin \alpha$$

α°	K	E	α°	K	E	α°	K	E
0	1.5708	1.5708	30	1.6858	1.4675	60	2.1565	1.2111
1	1.5709	1.5707	31	1.6941	1.4608	61	2.1842	1.2015
2	1.5713	1.5703	32	1.7028	1.4539	62	2.2132	1.1920
3	1.5719	1.5697	33	1.7119	1.4469	63	2.2435	1.1826
4	1.5727	1.5689	34	1.7214	1.4397	64	2.2754	1.1732
5	1.5738	1.5678	35	1.7312	1.4323	65	2.3088	1.1638
6	1.5751	1.5665	36	1.7415	1.4248	66	2.3439	1.1545
7	1.5767	1.5649	37	1.7522	1.4171	67	2.3809	1.1453
8	1.5785	1.5632	38	1.7633	1.4092	68	2.4198	1.1362
9	1.5805	1.5611	39	1.7748	1.4013	69	2.4610	1.1272
10	1.5828	1.5589	40	1.7868	1.3931	70	2.5046	1.1184
11	1.5854	1.5564	41	1.7992	1.3849	71	2.5507	1.1096
12	1.5882	1.5537	42	1.8122	1.3765	72	2.5998	1.1011
13	1.5913	1.5507	43	1.8256	1.3680	73	2.6521	1.0927
14	1.5946	1.5476	44	1.8396	1.3594	74	2.7081	1.0844
15	1.5981	1.5442	45	1.8541	1.3506	75	2.7681	1.0764
16	1.6020	1.5405	46	1.8691	1.3418	76	2.8327	1.0686
17	1.6061	1.5367	47	1.8848	1.3329	77	2.9026	1.0611
18	1.6105	1.5326	48	1.9011	1.3238	78	2.9786	1.0538
19	1.6151	1.5283	49	1.9180	1.3147	79	3.0617	1.0468
20	1.6200	1.5238	50	1.9356	1.3055	80	3.1534	1.0401
21	1.6252	1.5191	51	1.9539	1.2963	81	3.2553	1.0338
22	1.6307	1.5141	52	1.9729	1.2870	82	3.3699	1.0278
23	1.6365	1.5090	53	1.9927	1.2776	83	3.5004	1.0223
24	1.6426	1.5037	54	2.0133	1.2681	84	3.6519	1.0172
25	1.6490	1.4981	55	2.0347	1.2587	85	3.8317	1.0127
26	1.6557	1.4924	56	2.0571	1.2492	86	4.0528	1.0086
27	1.6627	1.4864	57	2.0804	1.2397	87	4.3387	1.0053
28	1.6701	1.4803	58	2.1047	1.2301	88	4.7427	1.0026
29	1.6777	1.4740	59	2.1300	1.2206	89	5.4349	1.0008
30	1.6858	1.4675	60	2.1565	1.2111	90	∞	1.0000

The definitions of the complete elliptic integrals of the first kind, $K(k)$, and the second kind, $E(k)$, are as follows:

$$K(k) = \int_0^{\pi/2} \frac{d\varphi}{\sqrt{1 - k^2 \sin^2 \varphi}}$$

and

$$E(k) = \int_0^{\pi/2} \sqrt{1 - k^2 \sin^2 \varphi} \, d\varphi$$

Approximate Formulas for $E(k)$; $k = \sin \alpha$

$$E(k) = \left[1 + 1.464 (\cos \alpha)^{1.65} \right]^{1/2}$$

Accuracy: 0.1%

References: **Rawe 1970**; see also **Merkle 1973, Raju 1979**

$$E(k) = \frac{\pi}{2} \cdot \frac{1}{1 + \lambda} \cdot \frac{4 - 0.18\lambda^4}{4 - \lambda^2}$$

where $\lambda = \tan^2 \frac{\alpha}{2}$

Accuracy: 0.05%

Reference: **Tada 2000**

APPENDIX M

A TABLE OF GAMMA FUNCTION $\Gamma(x)$

x	$\Gamma(x)$	x	$\Gamma(x)$
1.000	1.00000	1.500	.88623
1.025	.98617	1.525	.88729
1.050	.97350	1.550	.88887
1.075	.96191	1.575	.89094
1.100	.95135	1.600	.89351
1.125	.94174	1.625	.89657
1.150	.93304	1.650	.90012
1.175	.92520	1.675	.90414
1.200	.91817	1.700	.90864
1.225	.91192	1.725	.91361
1.250	.90640	1.750	.91906
1.275	.90160	1.775	.92499
1.300	.89747	1.800	.93138
1.325	.89400	1.825	.93825
1.350	.89115	1.850	.94561
1.375	.88891	1.875	.95345
1.400	.88726	1.900	.96177
1.425	.88618	1.925	.97058
1.450	.88566	1.950	.97988
1.475	.88567	1.975	.98969
1.500	.88623	2.000	1.00000

Values of $\Gamma(x)$ for $x < 1$ and $x > 2$ are calculated by the following formula:

$$\Gamma(x) = \frac{\Gamma(x+1)}{x} \text{ or } \Gamma(x) = (x-1)\Gamma(x-1)$$

Examples:

$$1) \quad \Gamma(0.65) = \frac{\Gamma(1.65)}{0.65} = \frac{0.90012}{0.65} = 1.38480$$

$$\begin{aligned} 2) \quad \Gamma(3.65) &= 2.65 \times \Gamma(2.65) \\ &= 2.65 \times 1.65 \times \Gamma(1.65) \\ &= 2.65 \times 1.65 \times 0.90012 = 3.93577 \end{aligned}$$

PROPERTIES OF GAMMA FUNCTION $\Gamma(x)$

Definition:

$$\Gamma(x) = \begin{cases} \int_0^{\infty} e^{-t} t^{x-1} dt & \text{for } x \geq 0^* \\ \lim_{n \rightarrow \infty} \frac{n! n^{x-1}}{x(x+1)(x+2) \cdots (x+n-1)} & \text{for any } x^{**} \end{cases}$$

* when x is complex; $Re(x) > 0$

** real or complex

Basic Formulas:

$$\Gamma(x+1) = x\Gamma(x)$$

$$\Gamma(n) = (n-1)! \quad (n = \text{positive integer})$$

$$\Gamma(x)\Gamma(1-x) = \frac{\pi}{\sin \pi x}$$

$$\Gamma(x)\Gamma\left(x + \frac{1}{2}\right) = \frac{\sqrt{\pi}}{2^{2x-1}} \Gamma(2x)$$

$$\Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}, \quad \Gamma(1) = 1, \quad \Gamma\left(\frac{3}{2}\right) = \frac{\sqrt{\pi}}{2}$$

For crack problems, the following formulas are useful.

$$\begin{aligned} (1) \quad & \int_0^1 x^{2\alpha+1} (1-x^2)^\beta dx = \left(\frac{1}{2} \int_0^1 x^\alpha (1-x)^\beta dx \right) \\ & = \int_0^{\pi/2} (\sin \theta)^{2\alpha+1} (\cos \theta)^{2\beta+1} d\theta \\ & = \frac{\Gamma(\alpha+1)\Gamma(\beta+1)}{2\Gamma(\alpha+\beta+2)} \left(= \frac{1}{2} B(\alpha+1, \beta+1) \right) \\ & \left(= \frac{\alpha!\beta!}{2(\alpha+\beta+1)!} \text{ when } \alpha, \beta = \text{positive integers} \right) \end{aligned}$$

(2) Special cases: $2\alpha + 1 = \gamma$, $\beta = -\frac{1}{2}$

or $\alpha = -\frac{1}{2}$, $2\beta + 1 = \gamma$

$$\int_0^{\pi/2} (\sin \theta)^\gamma d\theta = \int_0^{\pi/2} (\cos \theta)^\gamma d\theta = \frac{\sqrt{\pi}}{2} \frac{\Gamma\left(\frac{\gamma+1}{2}\right)}{\Gamma\left(\frac{\gamma}{2} + 1\right)}$$

when $\gamma = n$ ($n =$ positive integer)

$$\left\{ \begin{array}{ll} = \frac{(n-1)(n-3)(n-5) \cdots 3.1}{n(n-2)(n-4) \cdots 4.2} \cdot \frac{\pi}{2} & (n = \text{even}) \\ = \frac{(n-1)(n-3)(n-5) \cdots 4.2}{n(n-2)(n-4) \cdots 3.1} & (n = \text{odd}) \end{array} \right.$$

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REFERENCE INDEX

A

Anderson 1993, 589, 590, 591
ASME 1972, 26
ASTM 1972, 62

B

Barenblatt 1962, 94, 134, 165, 166, 266, 267, 346-348, 432, 434, 497, 547, 564
Begley 1972, 615
Begley 1976, 622
Bentham 1972, 47, 52, 55, 114, 153, 193, 204, 219, 221, 222, 223, 224, 225, 226, 227, 228, 229, 230, 231, 232, 233, 234, 236, 247, 251, 256, 261, 264, 269, 355, 365, 390, 393, 395, 396, 399, 400, 401
Berezhnitskii 1966, 297
Bilby 1963, 438
Bilby 1968, 547
Bowie 1956, 289, 290
Bowie 1964a, 46
Bowie 1965, 52
Bowie 1970a, 42
Bowie 1970b, 237
Bowie 1973, 264
Bowie 1976a, 33
Bowie 1976b, 33
Brown 1966, 41, 47, 53, 56, 59
Bueckner 1958, 7
Bueckner 1960, 55
Bueckner 1965, 390
Bueckner 1966, 193
Bueckner 1970, 52, 55, 497, 500
Bueckner 1971, 52, 55, 497, 500
Bueckner 1972, 346, 371, 390, 497, 500

C

Collins 1962, 365
Cottrell 1953, 547
Creager 1967, 8

D

Doran 1969, 214
Dowling 1976, 622
Dugdale 1960, 438

E

Emery 1969, 52, 55, 58
Emery 1972, 52
Erdogan 1962, 83, 84, 100, 104, 128, 129, 130, 138, 140, 142, 143, 165, 166, 227
Erdogan 1966, 457
Erdogan 1969, 485, 486
Erdogan 1973, 231
Erdogan 1982, 382, 383, 392, 394, 398, 402-409
Ernst 1979, 619

F

Fama 1972, 478
Feddersen 1966, 41
Fichter 1967, 256, 262, 272, 274, 275, 277, 279, 281
Folias 1965, 485
Folias 1967, 478
Forman 1964, 42

G

Galin 1953, 168, 341, 343, 344, 369
Gilman 1959, 266, 267
Green 1950, 364, 385, 388
Griffith 1920, 7, 19, 124
Gross 1964, 52, 53
Gross 1965a, 55
Gross 1965b, 58
Gross 1966, 79
Gross 1967, 53, 56, 59
Gross 1970, 61
Gustafson 1976, 33

H

Harris 1967, 390
Harris 1972, 515, 516, 521
Harris 1997, 485
Hartranft 1973, 197, 199, 206, 410
Hasebe 1978, 304
Hasebe 1980, 305
Hasebe 1981, 306
Hasebe 1987, 307
Hutchinson 1968, 616
Hutchinson 1979, 617, 621

Hutchinson 1992, 417, 423

I

Inglis 1913, 19
Irwin 1948, 7
Irwin 1952, 7
Irwin 1957, 3, 7, 12, 41, 47, 87, 124, 138, 140, 170, 184, 185, 190, 266
Irwin 1958a, 20, 22, 82, 124, 138, 140, 185
Irwin 1958b, 193
Irwin 1960a, 193
Irwin 1960b, 16, 630
Irwin 1961, 390
Irwin 1962a, 24, 134
Irwin 1962b, 364, 385
Irwin 1963, 583
Irwin 1969, 434, 438, 452
Irwin 1971, 489
Isida 1956, 234
Isida 1962, 40
Isida 1965a, 40, 227, 233
Isida 1965b, 40
Isida 1970a, 230, 288, 310-312, 331, 332
Isida 1971a, 42, 227, 232, 256, 260, 272, 274, 283, 284, 286, 287
Isida 1971b, 232, 286, 287, 456
Isida 1972, 263
Isida 1973, 40, 167, 309
Isida 1974, 242

J

Johnson 1965, 80

K

Kamei 1974, 167, 261, 263, 289, 294, 295, 313
Kapp 1980, 65
Kassir 1966, 386, 387
Kassir 1973, 334
Kassir 1975, 340, 366, 367
Kaya 1980, 70, 71
Keer 1974, 425
Keer 1989, 425
Keer 1990, 425
Kitagaswa 1975, 326
Kobayashi 1964, 232
Kobayashi 1973, 26
Kobayashi 1976, 412, 413
Koiter 1956, 247
Koiter 1959, 170
Koiter 1961, 247
Koiter 1965a, 193

Koiter 1965b, 40, 41, 221

L

Lachenbruch 1961, 25, 193
Landes 1972, 615
Libatskii 1967, 238, 241
Love 1944, 547
Lubahn 1959, 390, 488

M

Markuzon 1961, 162
Mendelson 1972, 42
Merkle 1973, 410, 411, 636
Muskhelishvili 1933, 18, 314

N

Narendran 1982, 319, 320
Neal 1970, 298
Neuber 1937, 1958, 10, 112, 114, 116, 153, 204, 256, 264, 365, 377, 379, 381
Newman 1971, 66, 235, 289, 291, 292, 296, 299, 308, 309,
Newman 1979a, 64
Newman 1979b, 602
Newman 1981a, 412
Newman 1981b, 64
Nishitani 1969, 296
Nishitani 1971a, 193
Nishitani 1971b, 454
Nishitani 1973, 303, 461
Nishitani 1976, 264

O

Okamura 1976, 27
Orowan 1949, 7
Ouchterlony 1975, 195, 212, 213, 218, 326, 328
Ouchterlony 1976, 328

P

Paris 1955, 247, 269
Paris 1957, 7, 16, 43, 50, 54, 57, 60, 67, 72, 73, 74, 134, 193, 494, 497, 583, 630
Paris 1960, 112, 114, 116, 247, 269, 497
Paris 1965, 12, 112, 114, 116, 124, 127, 138, 140, 142, 143, 390, 487, 513, 514, 602
Paris 1975a, 505
Paris 1975b, 33
Paris 1976, 497
Paris 1979, 621
Paris 1983a, 618, 621
Paris 1983b, 621

Paris 1988, 619
Peterson 1953, 10

R

Raju 1979, 410, 636
Rawe 1970, 636
Rice 1964, 15
Rice 1966, 623
Rice 1967, 247, 268, 269, 274, 283
Rice 1968a, 490
Rice 1968b, 616
Rice 1968c, 623, 634
Rice 1972, 497
Rice 1973, 619
Rice 1984, 623
Rich 1968, 458, 460
Riedel 1980, 622
Riedel 1989, 622
Ripling 1964, 489
Rooke 1969, 321
Rooke 1972, 242, 243
Rooke 1973a, 245
Rooke 1973b, 239

S

Sadowsky 1949, 364, 385
Sanders 1982, 471, 473, 475, 476
Sanders 1983, 473, 475, 476
Savin 1961, 10, 298
Savin 1968, 298
Saxena 1978, 63
Seeger 1973, 33, 156, 445
Segedin 1950, 363
Shah 1968, 389
Sih 1962a, 83, 128, 129, 130, 138, 140, 142, 143
Sih 1962b, 314, 497
Sih 1964, 112, 116, 138, 140, 142, 143, 165, 169, 227
Sih 1965a, 127, 289, 293, 318, 324, 325
Sih 1965b, 513, 514
Sih 1968, 334, 363, 367, 386, 387
Sire 1989, 245
Smith 1967, 410, 411
Sneddon 1946, 346
Sneddon 1951, 148, 346, 347, 377, 379, 381
Sneddon 1969a, 547, 561
Sneddon 1971a, 193
Sneddon 1971b, 42
Srawley 1964, 488
Srawley 1965, 488
Srawley 1967, 79

Srawley 1972, 61
Srawley 1976, 62
Stallybrass 1969, 321
Stallybrass 1970, 206
Stallybrass 1971, 221
Suo 1990a, 423, 426
Suo 1990b, 428
Swedlow 1972, 26

T

Tada 1970, 134, 176, 180
Tada 1971, 42
Tada 1972a, 42, 85, 86, 91, 92, 97, 98, 131, 132, 173, 174, 193, 221, 454, 456, 491, 583, 586, 589
Tada 1972b, 25, 42
Tada 1973, 41, 43, 47, 49, 53, 54, 56, 57, 59, 60, 67, 68, 72, 73, 74, 76, 77, 78, 80, 85, 86, 89, 91, 97, 98, 104, 110, 131, 132, 134, 145, 173, 174, 176, 180, 184, 190, 193, 195, 197, 202, 204, 206, 214, 221, 222, 223, 224, 225, 227, 228, 239, 240, 241, 244, 247, 248, 249, 250, 252, 253, 254, 255, 256, 257, 258, 259, 260, 262, 264, 265, 267, 270, 278, 282, 283, 298, 299, 300, 302, 303, 343, 344, 345, 351, 369, 370, 371, 373, 375
Tada 1974, 92, 95, 117, 121, 123, 148, 150, 162, 164, 168, 301, 341, 349, 350, 416, 418, 420, 422, 434, 435, 436, 437, 439, 440, 441, 442, 443, 444, 446, 447, 448, 449, 450, 451, 452, 453, 454, 455, 456, 457, 459, 462, 463, 464, 465, 466, 467
Tada 1975, 352, 353, 357, 358, 411, 413
Tada 1983a, 471, 473, 478, 485
Tada 1983b, 531, 532
Tada 1983c, 581, 586, 589
Tada 1985, 70, 71, 75, 88, 90, 93, 94, 99, 101, 103, 106, 107, 109, 111, 113, 114a, 126, 135, 137, 139, 141, 144, 146, 147, 149, 152, 158, 160, 172, 178, 182, 187, 189, 194, 195, 196, 198, 198, 199, 201, 203, 205, 206, 207, 208, 209, 210, 211, 213, 214, 215, 216, 217, 220, 226, 230, 231, 242, 263, 288, 289, 290, 291, 315, 316, 317, 321, 322, 323, 326, 327, 329, 330, 335, 336, 337, 338, 339, 340, 342, 343, 344, 345, 346, 347, 348, 349, 350, 352, 353, 354, 356, 357, 358, 359, 360, 361, 362, 366, 367, 372, 374, 376, 378, 380, 382, 383, 392, 394, 398, 400, 402, 403, 404, 405, 406, 407, 408, 409, 412, 414, 462, 471, 473, 475, 476, 486, 530, 531, 532, 533, 534, 535, 537, 538, 542, 543, 544, 545

Tada 1993, 547
Tada 1994, 547
Tada 1996, 561
Tada 2000, 79, 96, 118, 119, 154, 156, 191, 208,
209, 245, 304, 305, 306, 307, 416, 418, 420, 425,
429, 430, 479, 480, 482, 484, 536, 636
Terada 1976, 532
Tracey 1973, 410, 412
Tweed 1970, 162, 164, 547
Tweed 1972a, 237, 238
Tweed 1973, 244

U

Uflyand 1965, 334

V

Vazquez 1971, 581, 584, 586, 587

W

Watwood 1969, 26
Weinstein 1952, 365

Westergaard 1939, 3, 18, 19, 33, 35, 124, 547
Westmann 1964, 326
Westmann 1966, 388
Wigglesworth 1957, 193
Williams 1957, 22, 82
Williams 1971, 326
Wilson 1969, 219
Wilson 1970, 219, 221
Wilson 1972, 26
Winne 1958, 112, 116
Wundt 1959, 390

Y

Yamamoto 1972, 42, 46, 52
Yamera 1966, 238
Yokobori 1965, 167
Yokobori 1972, 290, 294, 295, 313

Z

Zahn 1965, 390

SUBJECT INDEX

A

- Additivity of crack stress fields
 - and K values, 22–23
- Additivity principles, 23
 - also see* superposition, Green's function method
- Airy stress functions, 17, 28, 31
- Algebraic sum of K, 13
 - also see* superposition
- Alternate expressions
 - for crack-tip stress fields, 6
- Alternating methods, 197, 199, 206, 322, 323, 410–413
 - see also* successive boundary stress correction method, successive stress adjustment (relaxation)
- Angle of twist, 640
- Anisotropic
 - elastic bodies, 13, 633,
 - homogeneous (rectilinear), 513
 - linear-elastic crack-tip stress fields, 513–514
 - materials, 21
- Anisotropy, 10
- Anti-plane shear, 623
 - see also* Mode III, pure shear
- Arc-shaped specimen, 65
 - ASTM E-399 standard specimens, 65
- ASME Nuclear Pressure Vessel Code, 601
- ASTM E-399, 62, 65, 489
- Asymptotic approximations, 41, 47, 52, 55, 234, 236, 256, 261, 390, 393, 395, 396, 399, 401
- Asymptotic interpolations, 67, 68, 72–74, 76–78, 215, 239, 241, 244, 260, 264, 265, 278, 282, 283, 329, 330, 400

B

- Beam (bending) theory, 231, 266, 267, 272, 275, 282
- Bend angles, 475, 476
 - also see* rotations, kinks
- Bend specimens
 - see* single edge cracked specimen, pure bending specimen, three-point bend specimen

- Blunting of crack, 8, 15
- Body force method, 264, 296
- Body force problems, 505
- Boundary collocation methods, 23–24, 42, 52, 53, 55, 56, 58, 59, 61, 63–66, 79, 219, 221, 232, 235, 287, 289, 291, 292, 296, 298, 299, 308, 309
- Boundary collocation points, 24
- Boundary conditions, 20, 23, 24, 625–62
- Bounding techniques, 603
- Boussinesq-Papkovich potentials, 371
- Bueckner displacement fields, 502, 505
- Bueckner-type singularities, 500–502, 508, 509, 512
- Burgers vector, 549

C

- C^* (time dependent J), 622
- Cantilever-beam specimen (or configuration), 15, 489
- Castigliano's theorem, 494
- Cauchy-Riemann equations, 18
- Center cracked test specimen, 40–45
- Charpy impact test, 620
- Circular cracks, 598, 599
- Circular holes, 294, 295, 313, 609
- Clip gages, 495
- Closed-form solutions, 17, 23
- Collinear dislocations, 550, 579
- Compact tension test specimen, 61–63
 - ASTM E-399 standard specimen, 62
- Compatibility, 490
 - strain, 17
 - stress, 490
- Compatibility equations, 27, 623, 625
- Complete elliptic integrals
 - of the first kind, 384, 635
 - of the second kind, 384, 600, 635
 - approximate formulas, 636
 - definitions of, 636
 - Table of, 635
- Complete shell solutions, 475, 476
- Complex potentials, 167, 313
 - expansions of, 40, 230, 256, 260, 263

- Complex stress potentials
 expansions of, 227, 232, 233, 272, 274,
 282, 283, 286–288, 309–312, 331, 332,
 456
- Compliance, 487
 effective, 492
 elastic, 11
 nonlinear, 614, 615, 618, 621
- Compliance calibrations, 15, 26, 42, 487–492
 experimental (by finite element
 computations), 26
- Compliance methods, 42
- Concentrated displacements, 548, 552, 553
also see nucleus of displacement
- Concentrated force K solutions, 494
also see splitting forces, wedge forces,
 Green's functions
- Configuration effects, 595
- Configuration functions, 593
- Conformal mapping, 293–295, 304–306,
 314, 316, 318, 324–326, 329, 330
- Conic section, 9
- Conjugate function method, 80
- Conjugate (pairs of) roots, 513, 514
- Constant (or fixed)
 displacements, 614
 loads (forces), 493, 494
also see fixity
- Constraint against contraction, 14
- Constraint conditions, 598
- Contour integral, 490
 path-independent, 490, 611
- Corrosion cracking, 2
- Cracks, 2, *and* throughout the book
 circular, 598, 599
 displacement-prescribed, 547, 562, 564, 579
also see dislocations
 edge, 594–596
 elliptical, 599, 600
 emanating from hole, 609–610
 fatigue
see fatigue cracking
 in residual stress fields, 529–545
 leading edge of, 2, 22
 Modes I, II, III, 2, *and* throughout the
 book
 part-through surface, 601–602
 penny-shaped, 598
 stop drilled, 8
 stress (or traction)-prescribed, 547, 550,
 562, 573, 578, 579
 through-the-thickness, 15
 tunnel, 598, 599
- Crack-absent stress distributions, 28, 529,
 562, 563
also see no-crack stress fields
- Crack arresters, 515
- Crack-arresting effects, 516
- Crack closure derivation of G , 12, 15
- Crack extension, 11, 13, 17, 22
 force, 11, 487
 processes, 15
 resistance curves, 587
also see J-R curves, R-curves
- Crack front, 11, 12
- Crack growth
see creep, J-controlled, fatigue
- Crack growth rates
 cyclic, 622
 fatigue, 622
 subcritical, 622
- Cracking
 corrosion, 2
 fatigue, 2, 622
- Crack length
see crack size
- Crack openings, 43, 48, 53, 56, 59, 62, 178,
 391, 394, 398, 403, 405, 407, 409
- Crack opening areas, 36, 43, 70, 125, 135,
 136, 139, 141, 144, 146, 147, 149, 151,
 154, 156, 157, 159, 171, 177, 181, 186,
 188, 194, 196, 198, 200, 203, 205, 207,
 213–217, 326, 470, 472, 474, 476, 478–
 481, 483, 485, 486, 529–538
- Crack opening displacements, 31, 529
- Crack (opening) profiles, 33, 37, 88, 90, 93,
 99, 102, 105, 108, 111, 113, 115, 118,
 125, 135, 136, 139, 141, 146, 147, 154,
 157, 159, 171, 177, 181, 186, 188, 194,
 196, 213, 378, 564, 568, 572
- Crack opening shapes, 342, 344, 348
also see crack opening profiles
- Crack opening stretch, 15, 433, 491, 614
- Crack propagation, 2
- Crack sizes, 5, 25
 determinations by compliance, 491
 increment, 26
- Crack stability, 621
- Crack surfaces, 2, *and* throughout Part I and
 Appendices
- Crack surface interference
see surface interference

Crack surface contact, 33
 Crack-tips, 2, *and* throughout Part I and Appendices
 Crack-tip (deformation, displacement, strain, stress) fields, 2, *and* throughout Part I and Appendices
 alternate expressions of, 6–7
 for linear-elastic bodies, 2–7
 for power-hardening materials, 616
 Crack-tip stress field environments, 5
 Crack-tip stress (field) intensity factors, 4
 also see stress intensity factors
 Crack-tip plastic (or non-linear) zone, 16, 22
 Creep crack growth, 622
 C-shaped specimen
 see arc-shaped specimen
 Cuspidal ends or tips, 37

D

DCB specimen
 see double cantilever specimen
 Debonded fibers, 429, 430
 Defects, 2
 Deformation theory of plasticity, 612, 615, 617, 621, 622
 Delta function, 552
 also see unit impulse function
 Dimensional analysis, 593–594
 Discontinuities, 547, 551
 Discrete (force) systems, 13
 Disk-shaped compact specimen
 see round compact specimen
 Dislocations, 547–579
 and cracks, 547–579
 edge, 547, 549
 screw, 547,
 of arbitrary shapes, 552
 Dislocation theory, 438
 Dislocation distributions, 261, 263
 Displacements, 31, 43, 48, 49, 53, 54, 56, 59, 62, 64, 67, 72–74, 88, 93, 99, 102, 105, 108, 111, 113, 118, 125, 137, 139, 141, 146, 147, 158, 159, 171, 172, 177, 181, 186–188, 193, 194, 196, 198, 200, 202, 204, 207, 213–217, 219, 342–346, 348, 352, 354, 356–359, 372, 376, 378, 391, 474, 534–536
 Displacement discontinuities, 547, 550
 Displacement fields, 3, *and* throughout Part I and Appendices
 Displacement measurements, 17
 Displacement modes, 3
 also see Modes

Distributed boundary tractions, 11
 Distributed force systems, 13
 Distributions of stresses, 2
 also see stress distributions
 Disturbance of crack, 8
 Double cantilever (beam) specimen, 15, 489
 Double edge cracked strip, 31, 596
 Double edge notch test specimen, 46–51
 Ductile materials, 581
 Dugdale(-Barenblatt) model, 432, 491

E

Edge dislocations, 547, 549
 Edge-sliding mode
 see Mode II
 Effective crack size, 16, 491, 583, 587, 630
 see plastic zone size (r_p) correction
 Effective compliance, 492
 Elastic closure, 12
 Elastic compliance, 11, 497
 Elastic energy, 11
 Elastic materials
 anisotropic, 21
 isotropic, 21
 orthotropic, 21
 Elastic-perfectly plastic, 616, 621, 623, 625
 Elastic-plastic boundary, 625, 627
 Elastic-plastic fracture mechanics, 15, 611
 Elastic-plastic models, 492
 analytical, 492
 Elasto-plastic pure shear analysis, 623–634
 Electrical potential calibration, 80
 Element sizes, 26
 Elevations of stresses, 2
 Elliptic integrals
 see complete elliptic integrals
 Elliptical cracks, 384–388, 599, 600
 Elliptical holes, 8, 10
 Elongation, 398
 End radius of notches, 8
 Energy balance, 11, 247, 268, 269, 283, 416, 418, 420, 422, 423, 426, 428–430
 Energy disappearance
 at crack tip singularity, 490
 Energy rate (or balance) analysis
 of crack extension, 11–12
 of effects of flaws, 11
 Equivalence
 between J and G , 491
 Equilibrium, 16
 Equilibrium equations, 17, 27, 624, 625

Extension, 470

F

Fatigue crack growth, 622
 under cyclic plastic conditions, 622
 Fatigue cracking, 2, 622
 Fiber pullout problem, 429
 Finite element mesh, 26
 Finite element methods, 25–26, 42, 46, 52,
 410, 412, 500, 505, 611
 Fixed displacements or loads
 see constant displacements or loads
 Fixity of displacement, 615
 Flaws, 3, 11
 Flaw-size effects in fracture, 5
 Flux of energy, 490
 Fourier series expansions, 366, 367
 Fourier transform, 42, 148, 247, 251, 256,
 262, 269, 272, 274, 275, 277, 279, 281,
 388, 561
 Fracture, 2, 5
 Fracture mechanics, 14, 15, 547
 see also linear-elastic fracture mechanics,
 elastic-plastic fracture mechanics
 Fracture mechanics stress analysis, 15
 Fracture process(es), 5,
 Fracture process zone, 15, 22, 491
 Fracture surfaces, 22
 Fracture toughness
 analysis, 587
 instability, 587
 Free boundary conditions, 20, 23
 Free-surface effects, 595

G

G , 11, *and* throughout the book
 as energy made available for crack
 extension, 11
 as generalized force, 11
 as point quantity, 11
 see crack extension force, total energy rate
 Gamma function, 117, 148–151, 206, 207,
 349
 definition, 637
 properties of, 638–639
 Table of, 637
 Generalized force, 11
 Generalized Hooke's law, 513
 Generalized plane stress, 14
 also see plane stress
 Green-Gauss theorem, 612

Green's functions, 497, 515, 529, 548, 552,
 553, 555, 562, 563, 572, 576–579,
 610
 also see concentrated force solutions,
 splitting forces, wedge forces
 Green's function methods, 13, 23, 52, 55, 58,
 598
 also see integrations
 Griffith crack, 23, 560, 561, 569, 573
 Griffith crack problem, 21
 Griffith energy rate, 497, 612
 Griffith theory, 11
 Griffith-Irwin theory, 11

H

Hankel transform, 355
 Higher-order terms, 3, 4
 History dependence, 622
 Hodograph method, 624
 Holes
 circular, 294, 295, 313, 609
 elliptical, 8, 10
 Hooke's laws, 14, 17
 for anisotropic material, 513
 generalized, 513
 HRR (crack tip) plastic field equations, 616–
 618
 Hypergeometric series, 561

I

I, II, III (subscripts), throughout the book
 see Modes I, II, III
 Imaginary part, 18, 19, 20, 549
 In-plane bending, 31, 33, 37, 445
 Influence function, 245
 Input work, 490
 Integral equations, 24, 42, 162, 164, 193,
 206, 214, 221, 224, 231, 239, 241, 244,
 296, 323, 326, 402–409, 423, 428, 478,
 486, 547, 561
 Integral transform, 193, 195, 204, 212–214,
 218, 219, 221, 237, 242, 245, 326, 328,
 342, 346, 347, 355, 385, 392, 394, 398,
 402–409, 425, 485
 Integrations, 89, 91–93, 107, 109–111, 117,
 118, 130, 142, 146, 148–153, 158, 190,
 191, 199, 201, 203, 206–208, 210, 211,
 335–340, 345, 347–357, 360, 361, 370,
 373–375, 515, 529–538, 542–545, 548,
 553, 554, 561–563, 572, 573, 577, 578,
 597, 610

also see Green's function method, superposition

Intensitiy

- of linear-elastic stress distribution, 4
- of load transmittal, 4
- of local stress field, 2
- of stress(-strain) singularity, 620

Interactions between modes, 12

Interference of closure, 33

Internal cracks, 33

Invariants, 21

Isotropic materials, 21

J

J-controlled crack growth, 617, 619, 621

J-integral, 26, 491, 611–622

- as elastic-plastic analog to G , 15
- field equation interpretation of, 616
- for compact tension specimen, 619
- for double cantilever beam specimen, 620
- nonlinear compliance definition of, 614–616
- as nonlinear counterpart of G , 11
- physical interpretation of, 612
- time rate dependent, 622

J(-integral)-R curves, 617, 621

also see crack extension resistance curves, R-curves

K

K , K_I , K_{II} , K_{III} , 4, *and* throughout the book
see stress intensity factors

as fracture correlation parameters, 5

K estimates

- engineering estimates, 593–610
- from finite element methods, 25–26

Kinks (at cracked section)

see rotations, bend angles

L

Laplace's equation, 625, 629

Large-scale yielding, 16, 433, 587

Laurent's expansion

see complex (stress) potentials

Leading edge of crack, 3, 22

straight, 488

Leading terms of crack-tip stress fields, 4

Least squares fitting, 34, 41, 47, 53, 56

LEFM, 581, 583, 586, 587, 589, 591

also see linear-elastic fracture mechanics

Length of closed portion, 33

Length of closure, 34, 569, 571

Length of contact, 564, 565, 567, 568

Length of normal, 384, 600

Linear-elastic

- analysis, 16
- behavior, 17, 487
- crack-tip (stress) fields, 2, *and*
throughout Part I and Appendices
- fracture mechanics, 14, 15, 16, 581, 630
- stress analysis, 2, *and* throughout Part I
and Appendices
- stress distributions, 4
- stress fields, 2, *and* throughout Part I and
Appendices

Load paths, 4

redistribution of, 4

Load (point) displacements, 11, 12, 26, 487, 497,

Load transmittal, 4

Loading (force) systems, 13

Local elevations of stresses, 2

Local stress fields, 4

Local(ized) yielding, 583

M

Mapping, 624, 625, 627–630

Mapping collocation method (technique),
237, 298

Mapping function methods, 46, 52, 264,
289, 290

Mathematical models, 16

Mellin transform, 239, 244

Modes

of crack surface displacement, 2

Modes I, II, III, 2, *and* throughout the book

Moments of inertia, 416, 418, 420, 422

Multiple integral equations, 547

Muskhelishvili's method, 83, 84, 100, 128–130,
162, 165, 166, 169, 227, 457, 458, 460

N

Negative K_I , 31

Negative openings, 31

also see surface overlapping

Neuber-Papkovitch potentials, 369, 389

No-crack stress fields, 23, 25

also see crack absent stress distributions

Nonlinear analyses, 16

Nonlinear compliance, 614, 615, 618, 621

also see compliance

Nonlinear effects, 2

Nonlinear elastic stress analysis, 611–613,

Nonlinear zone, 16
 Nonlinearity, 2, 4, 8, 583
 Non-self-equilibrating loadings, 514
 Non-small-scale yielding, 586
 Notches, 2
 slender, 8, 10
 Notched round bars, 488
 Notch tips, 8, 9
 Nucleus of displacement, 552
 Numerical integrations, 24

O

One function approach
 see Westergaard one function approach
 Openings, 90, 93, 108, 111, 125–135, 139,
 141, 144, 146, 147, 149, 151, 157, 159,
 171, 181, 186, 198, 201, 203, 205, 207,
 214–217, 342, 344–350, 352–354, 356–
 359, 361, 362, 462, 463, 531–538, 542–545
 Opening displacements, 64
 see also crack openings
 Opening displacements
 of crack surfaces, 494–495
 near crack tip, 495
 Opening mode
 see Mode I
 Opposing semi-infinite cracks, 33
 Orthotropic materials, 21
 Overlapping, 31, 32, 154
 also see surface overlapping, surface
 interference

P

Papkovitch-Neuber potentials, 334, 341, 344
 Parabola, 9, 35
 Parabolic arc cracks, 320
 Parameters, 2, 4
 Parametric angle, 384, 600
 Paris' equation, 43, 50, 54, 57, 60, 67, 70, 72,
 73, 74, 99, 103, 106, 113, 115, 118, 126,
 135, 139, 144, 146, 149, 160, 172, 193–
 196, 198, 201–203, 207, 213–215, 342,
 343, 348, 349, 352, 372, 374, 376, 378,
 380, 392, 394, 398, 400, 471, 473, 475,
 476, 478, 479, 482, 484–486, 529–538
 Paris' formula, 602
 Partly closed cracks, 31, 32
 Path-independence, 490, 491
 Path-independent (contour) integral, 611, 612
 Penny-shaped cracks, 598
 also see circular cracks

Periodic (array of) cracks, 170–192
 Periodic (array of) dislocations, 551, 554, 555, 558,
 559
 Physical plane, 624–632
 Piecewise continuous, 553, 554
 Pinching forces (loads), 511, 515–518
 crack arresting effects of, 516
 Plane-extensional problems, 3
 Plane strain (conditions), 3, 14, *and*
 throughout the book
 mathematical definitions of, 14
 Plane strain and plane stress
 for fracture mechanics purposes, 14–15
 in fracture mechanics terminology, 14
 Plane strain constraint
 see constraint against contraction
 Plane strain fracture, 14, 15
 Plane stress (conditions), 4, 14, *and*
 throughout the book
 mathematical definitions, 14
 Plane stress fracture, 14
 Plastic collapse loads, 581, 589–591
 Plastic design, 621
 Plastic limit loads, 581
 Plastic slips
 see slips
 Plastic zone, 14, 15, 16, 625, 626
 Plastic zone instability, 581, 582
 analysis, 581–592
 concept, 581–592
 failure criterion, 581
 Plastic zone size, 14, 15, 432, 433, 583
 correction (adjustment), 16, 581, 583, 630
 index, 16, 587
 also see effective crack size
 Plasticity, 2, 4, 8, 15
 also see nonlinear effects, nonlinearity
 theory of, 14
 Point of separation (loss of contact)
 of surfaces, 33, 35, 556, 557, 562, 564,
 565, 567
 Point quantity
 G as a, 12
 Poisson's ratio, 4
 Polar moment of inertia, 430
 Positive branch, 24
 Potential energy change, 11
 Power hardening, 634
 Power series, 495
 Pressure vessels, 581, 582, 589
 Principal stresses, 21

Profile of crack
 see crack opening profile
 Progressive separational process, 17
 Proportional loading, 37
 Proportional straining, 617
 Punch problems, 168
 Pure bending specimen, 55–57
 Pure shear, 3, 513, 514, 623
 Pyramid-shaped distribution, 359
 PZI
 see plastic zone instability

R

Ramberg-Osgood relation, 617
 R-curves, 618, 621
 also see crack extension resistance curves, J-R curves
 Real part, 18, 19, 20, 549
 Reciprocal theorem, 497, 499
 Reciprocity, 88, 99, 103, 106, 113, 139, 178, 182, 187, 342–344, 376, 513
 Redistribution of stresses
 due to cracks or notches, 2
 Redistribution of load paths, 4
 Relationships between G and K , 12–13, 490–491, 493
 Relative displacements
 see displacements
 Relative rotations, 496
 of arms, 620
 also see rotations
 Residual stresses, 505, 529
 Rice's analysis
 for pure bending, 618–620,
 Rice's J-integral, 611
 also see J-integral
 Rigid wedges
 see wedges
 Ring load, 464
 Roof-shaped distribution, 358
 Rotating discs, 243, 246
 Rotations, 70, 103, 109, 115, 219, 220, 380, 394, 400, 471, 473, 537, 538
 Round compact specimen, 64
 Rules for superposition for K and G , 13
 r_y – correction
 see plastic zone size correction

S

Sanders' solutions, 471, 473
 also see complete shell solutions

Self-balanced (equilibrating) stresses, 488, 489, 594–596
 Semicubical parabola, 35
 Semi-elliptical arc cracks, 320
 Semi-major axis, 9
 Semi-minor axis, 10
 Sensitivity to flaws, 2
 Separations, 15
 Separation profile, 35
 Series expansion, 495
 also see complex (stress) potentials
 Shells
 cylindrical, 581, 582
 spherical, 581, 582
 Shell parameter, 582
 Simple beam theory, 416, 418, 420, 422, 489
 also see beam (bending) theory
 Simple radial stresses, 226, 229
 Single edge cracked strip, 31, 33, 505, 507
 Single edge notch test specimen, 23, 52–54
 Single stress function approach, 20
 also see Westergaard one function approach
 Singular integral equations, 55, 70, 71, 231, 382, 383, 390
 Singularities
 Bueckner-type, 500–502
 inverse square root, 4, 22
 strength of, 4
 Skew-symmetric fields, 2, 3, 503
 Slender ellipses, 10
 Slender notches, 8–10
 and stress concentrations, 8–11
 Slips, 625, 626, 633
 Small scale yielding, 2, 5, 14, 15, 16, 432–434, 583, 622, 629
 effects of, on linear-elastic fracture mechanics, 16
 Smooth separation of surfaces, 36
 Software, 676
 Solid mechanics, 14
 Splitting forces, 22, 529
 also see concentrated forces, Green's functions, wedge forces,
 Spot weldings, 515
 Stability analysis, 581
 Stamp problems, 377, 379, 381
 Stiffness matrix, 26
 Strain compatibility, 17
 also see compatibility equations
 Strain energy, 11, 15, 487, 493, 494
 density, 490
 reduction of, 11

Strain-path independent, 490
 Stress analysis, 2, *and* throughout Part I and Appendices
 Stress compatibility, 490
 also see equilibrium equations
 Stress concentrations, 8–10
 from stress intensity factors, 8–11
 Stress concentration factors, 112, 114, 124, 153, 204, 234, 256, 264, 365, 377, 379, 381
 Stress concentration theory, 10
 Stress distributions, 2, *and* throughout Part I and Appendices
 surrounding a crack, 4
 Stress fields, 2, *and* throughout Part I and Appendices
 Stress field energy, 25
 density, 21
 Stress-free boundaries, 2
 Stress functions, 19, 21, 624–629
 see also Airy stress function, Westergaard stress functions
 Stress function methods, 17–22
 Stress function surface, 632
 Stress intensity factors (K , K_I , K_{II} , K_{III}), 4, *and* throughout the book
 engineering estimates of, 593–610
 Stress plane, 624–632
 Stress relaxation, 303
 Stress removal, 24
 Stress-strain relations, 27
 Stress(-strain) singularity, 4, 33, 583
 intensity of, 620
 inverse square root, 22
 Strip yield models, 431–467, 491, 614
 Strip yield model analysis, 36, 432–433
 St. Venant's principle, 596
 Successive boundary stress correction
 method, 24–25,
 Successive stress adjustment, 197, 456
 Successive stress relaxation, 193, 454
 Superposition, 13, 33, 34, 37, 95, 97, 104, 127, 132, 140, 141, 147, 160, 174, 202, 204, 205, 209, 216, 222, 223, 315, 317, 358, 359, 362, 364, 386, 414–444, 446–455, 459, 462–467, 505, 515, 529, 536, 550, 555, 558, 562, 569, 573, 576, 594, 596
 Superposition method, 33
 Superposition of G and K results, 13
 Surface contact, 33

Surface cracks, 600–608
 almost semi-elliptical, 602–604
 semi-elliptical, 600–604
 irregular, 604–607
 part-through, 601
 Surface interference, 31, 33, 35, 155
 effect of, 31–38, 155
 Surface overlapping, 31, 33, 36
 Symmetric axis, 33
 Symmetric fields, 2, 3, 503
 System-isolated condition, 12, 487

T

Tearing instability theory, 618, 621
 Tearing mode
 see Mode III
 Theory of elasticity, 3, 14
 two-dimensional, 3
 Theory of plasticity, 14
 Thermal stress problems, 505
 Three-dimensional potential
 functions, 363, 385–388
 Three-dimensional problems, 2, *and* throughout Part I and Appendices
 Three-point bend test specimen, 58–60
 Through-wall cracks
 in shells, 581, 582, 587, 589
 circumferential, 581, 582, 590
 longitudinal, 581, 582, 583
 Total elastic energy, 11
 Total strain energy, 11
 Total energy rate (G), 12
 Traction-free surfaces, 28
 Transverse loads, 515
 also see pinching loads
 Two-dimensional problems
 plane-extensional (Modes I & II), 2
 pure shear (or torsion) (Mode III), 2
 and throughout the book
 Two-step approach (or process), 23
 also see Green's function method, superposition
 Tunnel cracks, 598, 599

U

Unified formulation for in-plane
 two-dimensional problems, 26–27
 Unit impulse function, 552
 also see delta function
 Unit step function, 552
 Unloading, 613, 622

Unrestricted plastic flow, 581, 589–591
Unstable crack growth, 581

V

Vertical displacements
 see displacements
Virtual
 displacements, 490
 forces, 493, 494
 increment of crack area, 487
Void growth and coalescence, 15
Volterra dislocations, 547–549
Volumes of crack, 342–350, 352–354, 356–
 362, 391, 397, 542–545

W

Wedges, 94, 95, 120, 122, 161, 163, 547,
 548, 557, 560–565, 568, 571, 573–575
Wedge forces, 445, 596, 599
 also see concentrated forces, Green's
 functions, splitting forces
Wedge force solutions, 596–598
 also see concentrated force solutions, Green's
 functions
Weight functions, 497–512
 closed-form, 503–504, 508–512

Weight function displacements, 502, 505,
 507, 510
Weight function method, 52, 55, 346, 497–512
Westergaard one function approach, 18
 completeness of, 27–31
Westergaard stress functions, 28, 31, 82, 85–
 91, 94–98, 101, 107, 109, 111, 114, 121,
 123, 124, 131, 132, 134, 137, 138, 140,
 154, 162, 164, 170, 172–174, 176, 178,
 180, 182, 184, 185, 187, 189, 193, 195,
 197, 199, 204, 221, 224, 226, 229, 247–
 249, 250, 253–259, 262, 264, 270, 434,
 438, 446, 452, 503, 534, 535, 547–579
Westergaard Z functions, 22
Work done in elastic closure, 12
Work-producing displacements, 499

Y

Yield criterion, 16
Yield surfaces, 625–629
Yield zone, 16
 shape of, 16

Z

Z functions, 21, 22, 24
 see Westergaard stress functions
Zone of plasticity, 14

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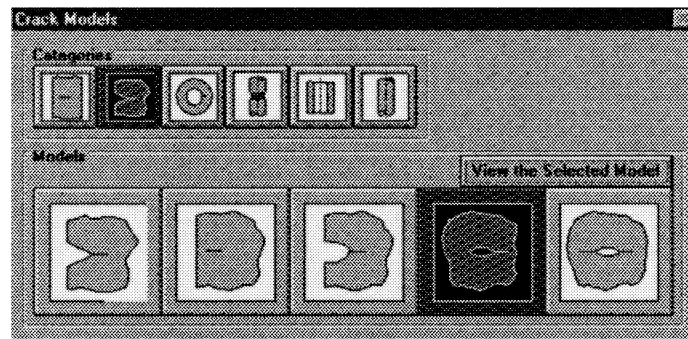
SOFTWARE FOR EVALUATION OF STRESS INTENSITY FACTORS BY USE OF FORMULAS IN THE STRESS ANALYSIS OF CRACKS HANDBOOK

The third edition of *The Stress Analysis of Cracks Handbook* contains many formulas and plots for evaluation of stress intensity factors. The use of such results is made more convenient by the software that is included in the accompanying computer diskette. The software is called SmartCrack-Lite, is in Windows, and contains the following 21 *K*-solutions included in the Handbook.

Standard Specimens	Infinite & Semi-Infinite	Round Bars
center cracked strip	crack from V-notch	exterior circum. crack
double edge crack	crack near edge	interior crack
single edge crack	crack from elliptical notch	Hollow Round Bars
3-point bend	crack from elliptical hole	interior crack
compact tension	2 cracks from ell. hole	exterior crack
arc shaped	Disks	Shells
round compact	edge loaded center crack	axial crack cylinder
	edge crack	circum. crack cylinder
		sphere

Many of the *K*-solutions include the influence function, which normally requires numerical integration for their use. The numerical integration procedure is included in the software and is transparent to the user. This makes application of the influence functions much easier. Further description of the software is included on the accompanying diskette, which also describes the installation procedure for SmartCrack-Lite. The following are the minimum system requirements to run SmartCrack-Lite: any IBM-compatible running Windows 3.1, 95, 98 or NT 4.0, a hard disk, and a 3 1/2" floppy disk.

The dialog box for the selection of cracks in the "infinite and semi-infinite planes" category is shown below as an example.



The *K*-solutions available in the software are a subset of the solutions included in a more comprehensive software package called SmartCrack, hence the name SmartCrack-Lite. The full SmartCrack software package, which is available from Engineering Mechanics Technology, Inc., in San Jose, California, contains many additional *K* solutions, and is able to perform fatigue and stress corrosion crack growth calculations, evaluate critical crack lengths, and evaluate crack opening areas. An extensive compilation of material fatigue crack growth characteristics is included.